



BUSINFO 701
Group 2

P R E S E N T A T I O N

Analysis Sustainable Development in Health Sector



World Health Statistics 2017:
Tracking Health and SDG Progress
Through Key Indicators





Who we are



We are a team of passionate analysts, dedicating to the study of global health and sustainable development.

Our presentation seeks to promote awareness and inspire collaborative action toward building sustainable and equitable health systems worldwide.





Bridging Gaps for Sustainable and Equitable Health Systems



Drawing on insights from the United Nations, the World Health Organization and World University Rankings statistics, we aim to highlight the key trends, progress, and challenges in achieving health-related Sustainable Development Goals (SDGs)

➤ Understanding the Factors

- Healthcare density impacts
- Regional disparities affect outcomes

➤ Analyze the Drivers

- Economic Indicators
- Educational Infrastructure

➤ Address Limitation

- In-place Systematic Monitoring and Escalation Process

➤ Call to Action





Methodology



✓ Data Sourcing

1. Country Statistics from the United Nations
2. World Health Statistics from the World Health Organization
3. World University Ranking
4. Real-time World Statistics from World Meter

✓ Data Wrangling

1. Clean and transform raw data to ensure accuracy and consistency
2. Select key attributes, including healthcare workforce densities, education levels and economic indicators
3. Merge through Python to create a unified dataset for cross-analysis.

✓ Visualization and Analysis

1. Develop visualization in Power BI and Python to assess the relationship between different indicators



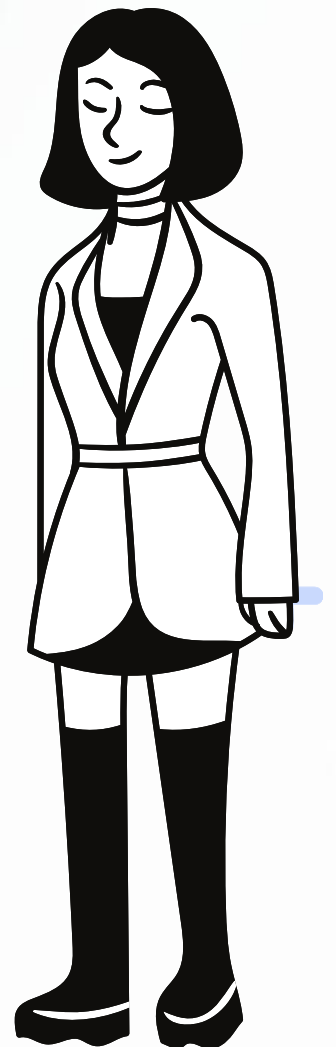


Understanding the Factors



How does the **density of medical staff** differ between **regions** and affect the **health and lifespan**?

Methodology: Correlation regression analysis

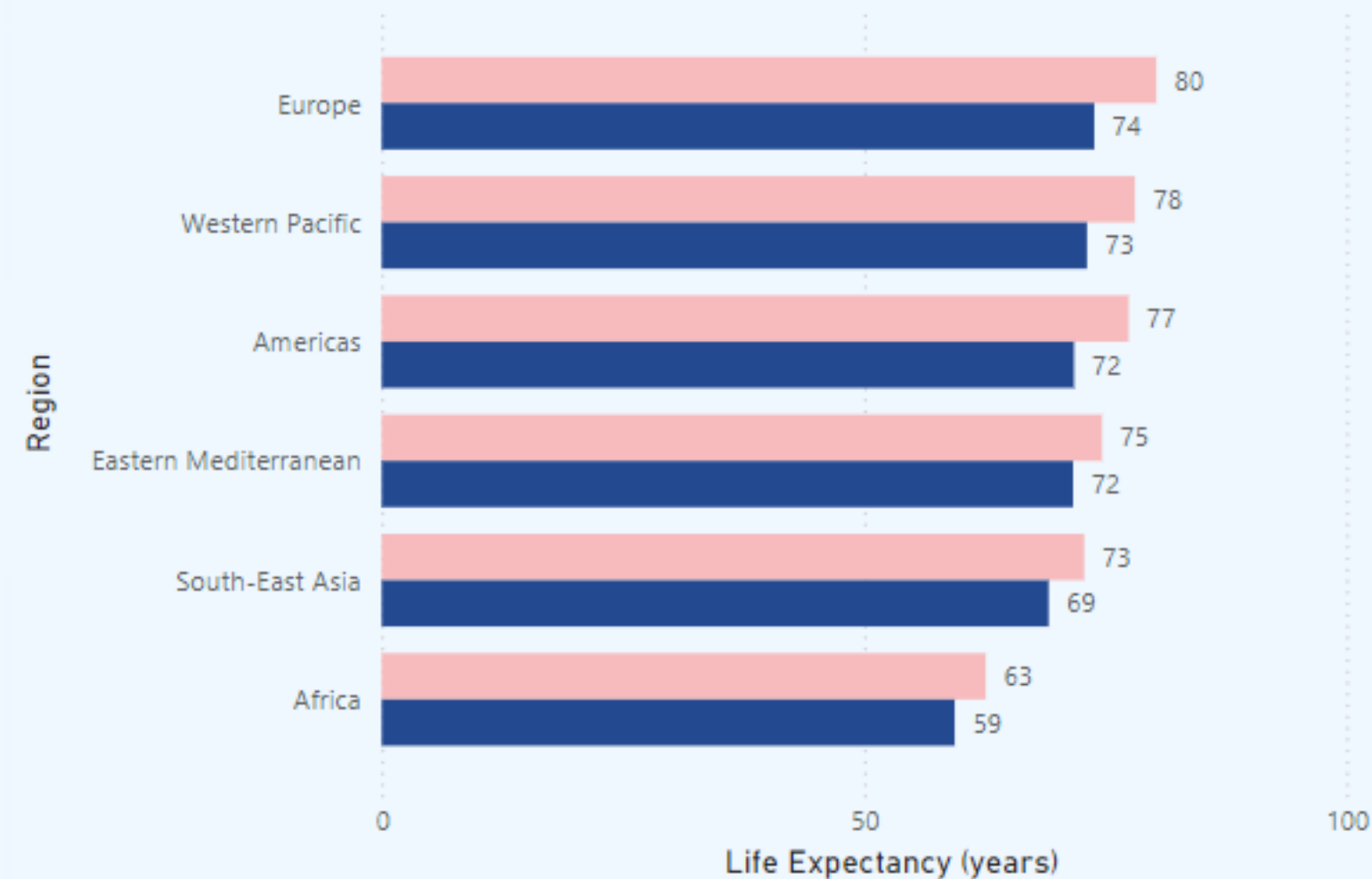


Region Difference of Medical Staff Density and Lifespan

- South-East Asia and Africa have less medical staff than world average.
- **Western Pacific** and **Europe** have the longest lifespan.
- The higher the medical staff density, the longer the lifespan.

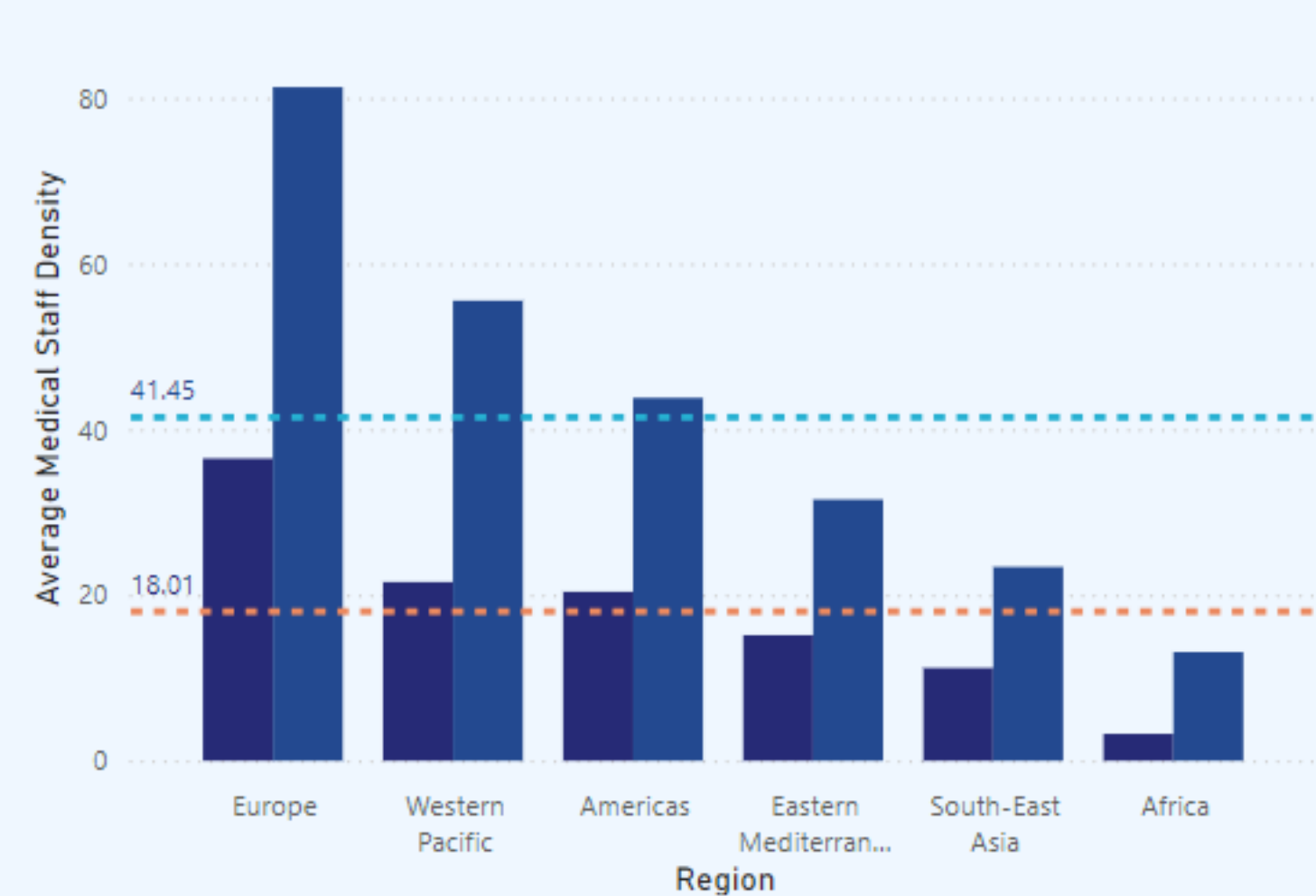
Average of Life Expectancy by Region

● Average of Female life expectancy ● Average of Male life expectancy



Average of Medical Staff Density by Region

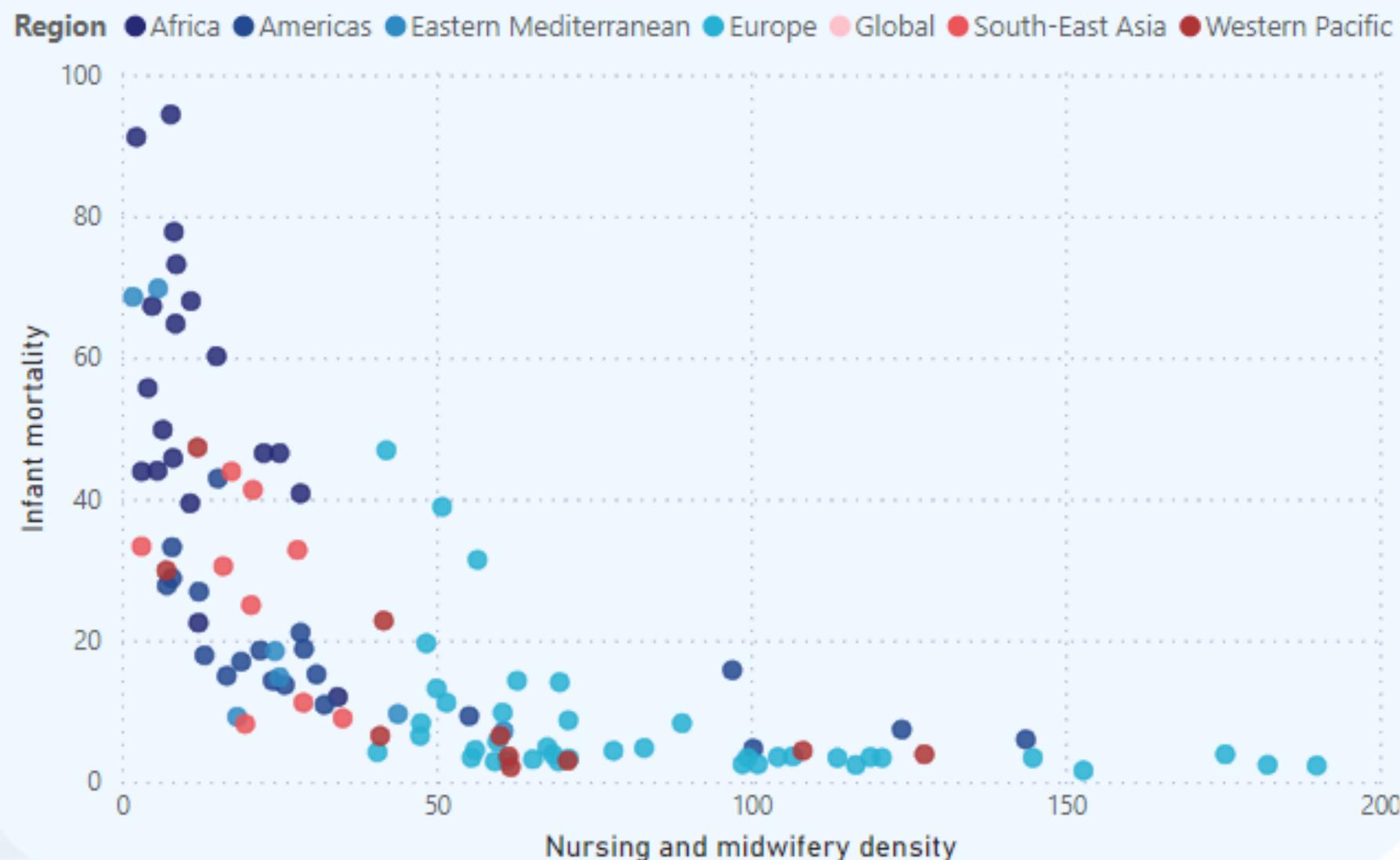
● Average of Medical doctors density ● Average of Nursing and midwifery density



Relationship Between Density of Medical Staff and Infant Mortality

- **European countries** show the lowest infant mortality.
- When doctor density goes up by 16.08 (per 10,000 people), infant mortality will decrease by 6.5%.
- When the Nursing and midwifery density goes up by 43.91 (per 10,000 people), the average infant mortality decreases by 11.22%.

Relationship of Nursing and Midwifery density on Infant Mortality by Region



Key influencers Top segments



What influences Infant mortality to ?

When...

Nursing and midwifery density goes up 43.91

Medical doctors density goes up 16.08

...the average of Infant mortality decreases by

11.22

6.5

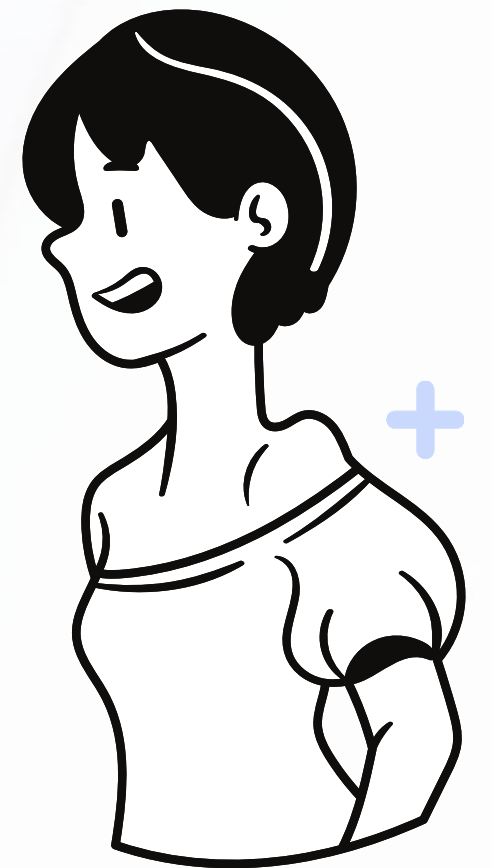


Analysing the Drivers



What is the Relationship Between the
Density of Different Medical Staff and
Economic Indicators Across Countries?

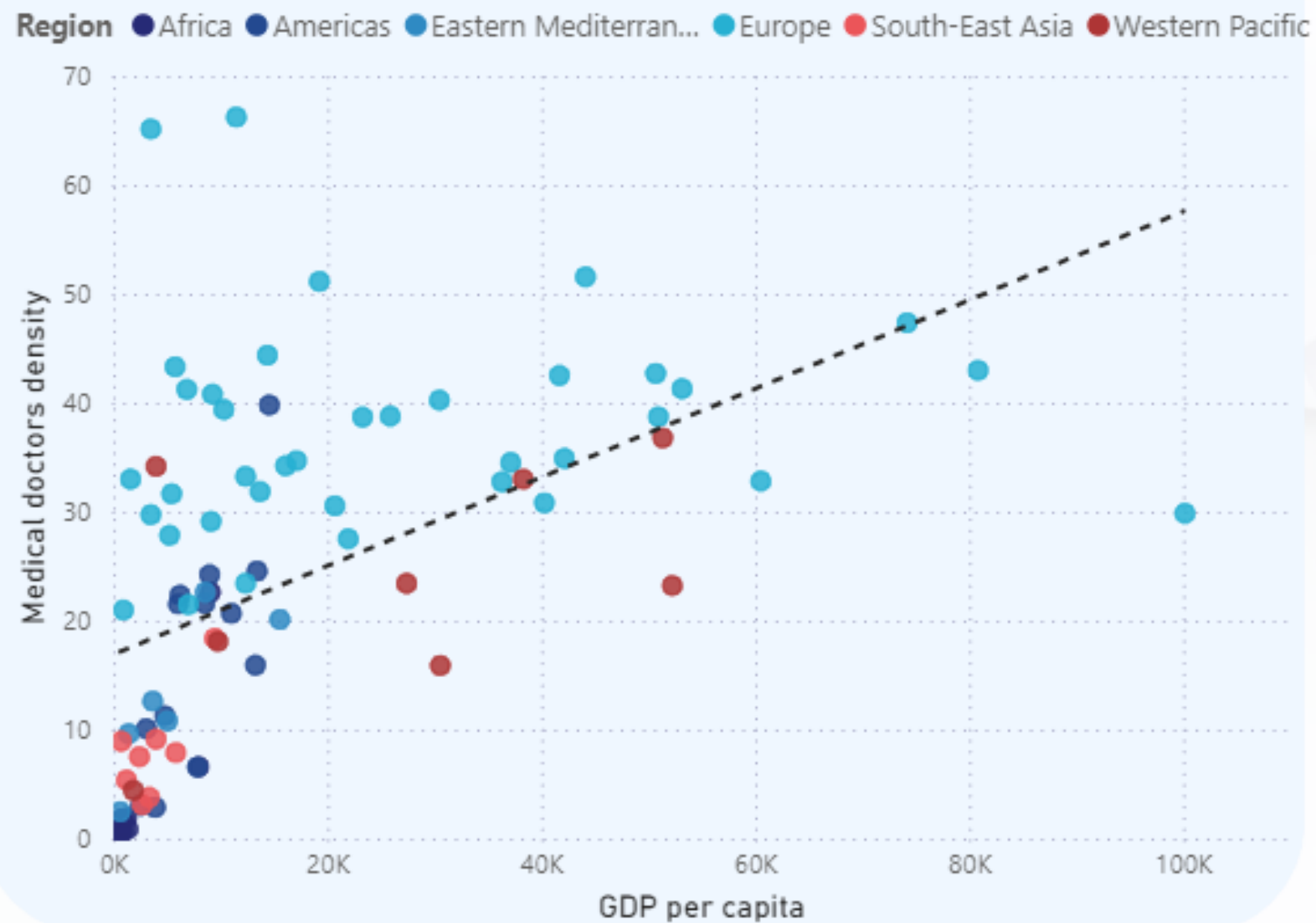
Methodology: Correlation regression analysis



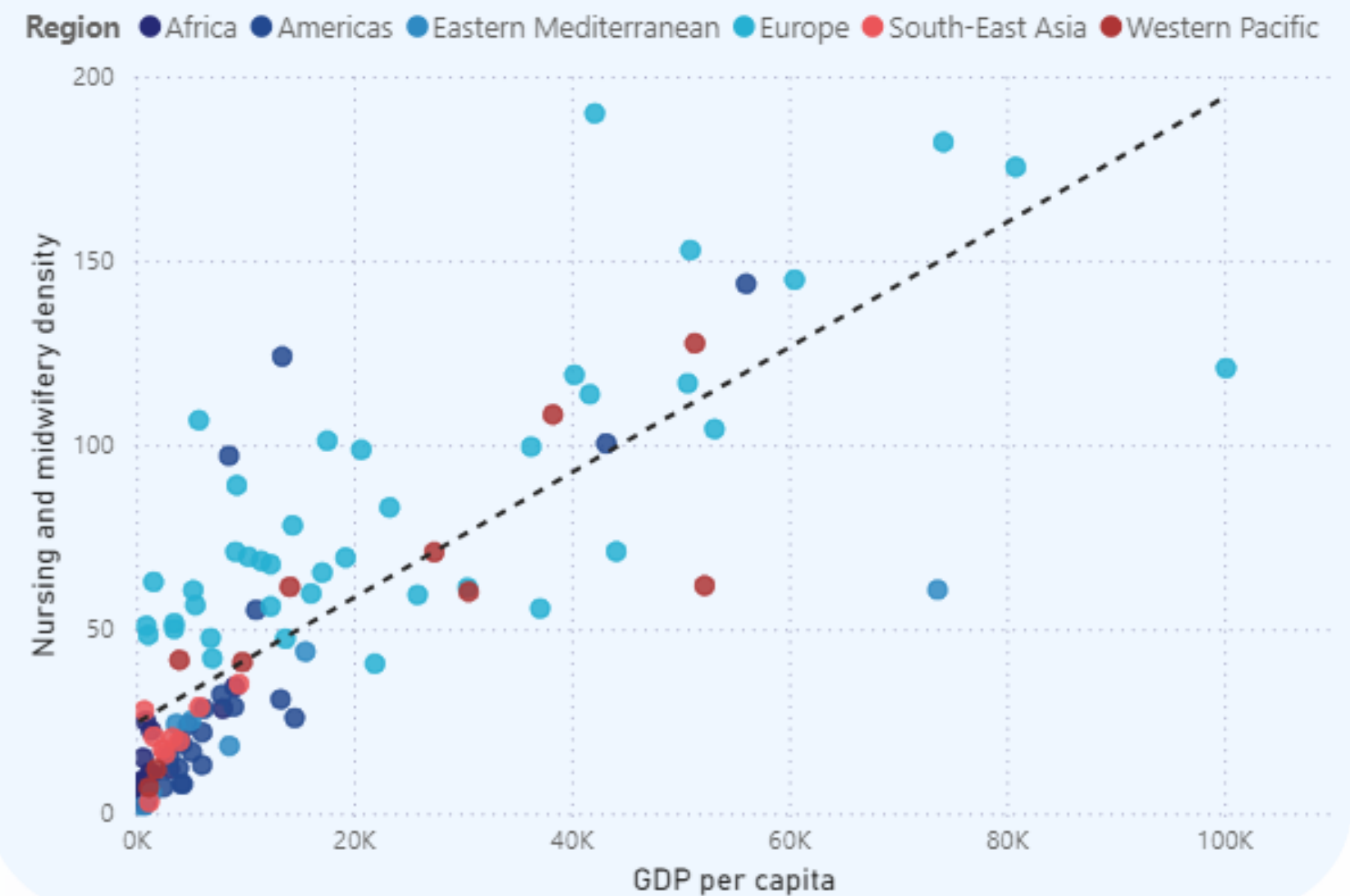
Relationship Between Density of Medical Staff and GDP Per Capita Across Regions

- GDP per capita has a **positive** relationship with both the density of medical staff
- **Regional** differences are also evident

Relationship of GDP Per Capita on Medical Doctors Density by Region



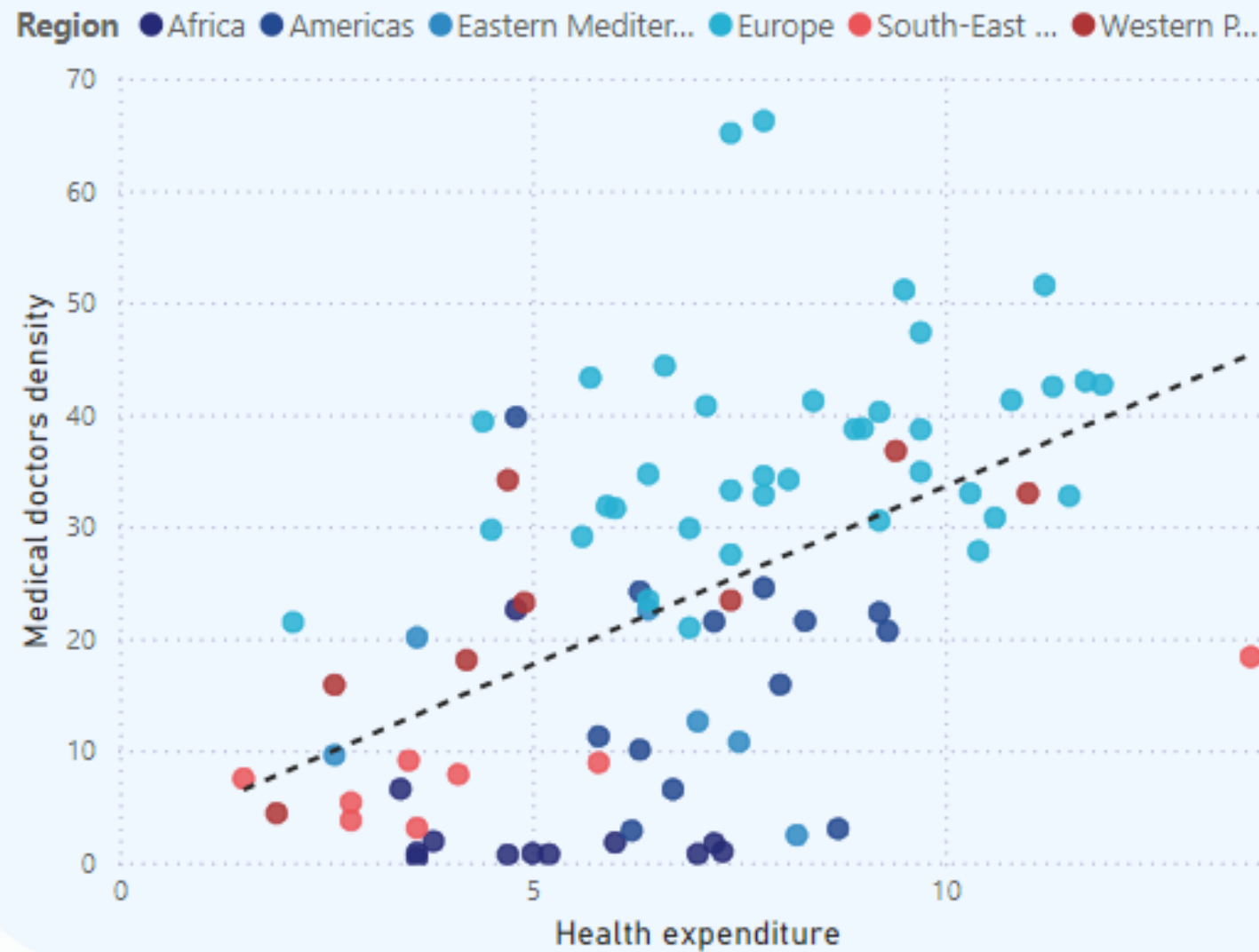
Relationship of GDP Per Capita on Nursing and Midwifery Density by Region



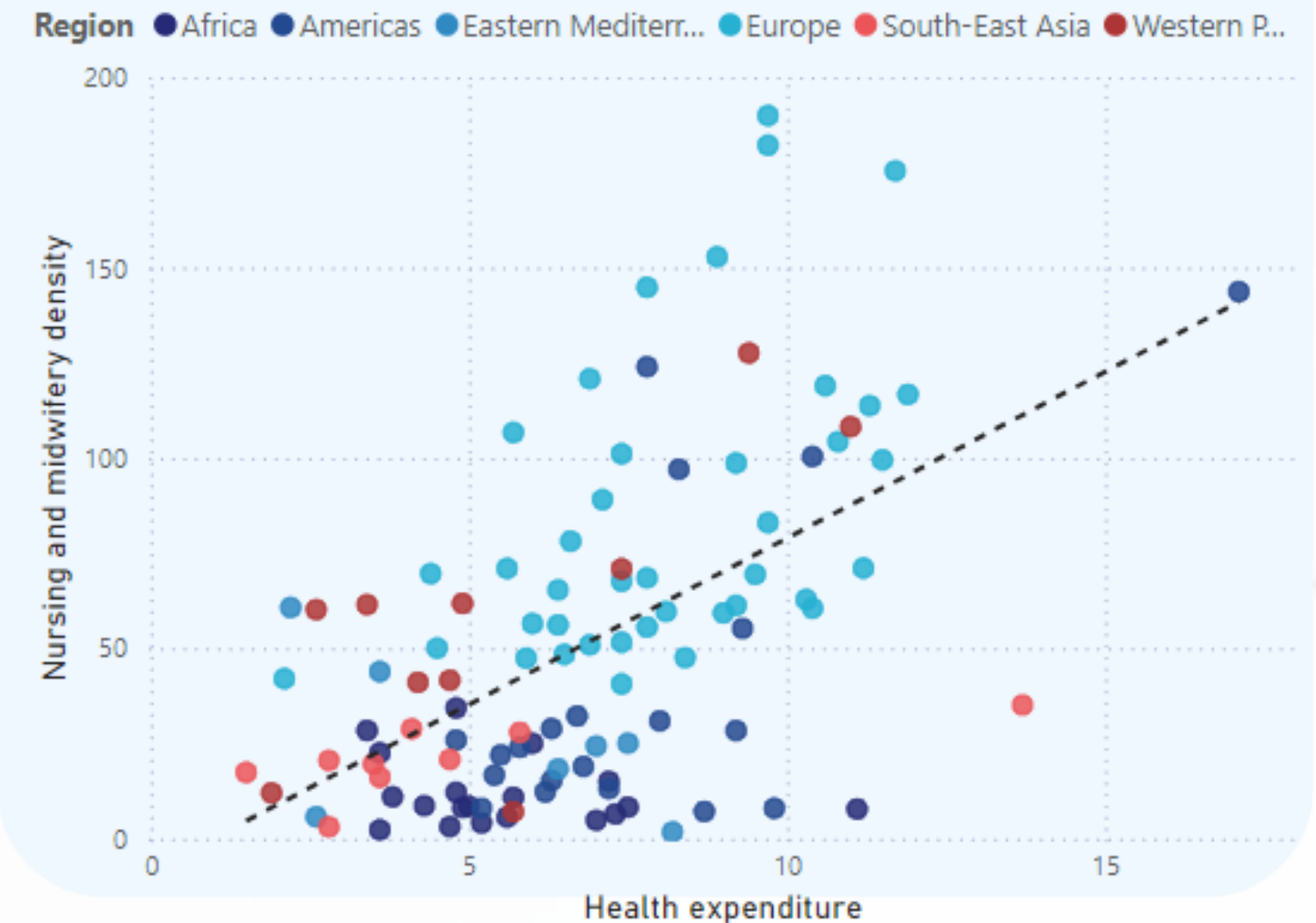
Relationship Between Density of Medical Staff and Health Expenditure Across Regions

- Health expenditure also has a **positive** relationship with both the density of medical staff and evident **regional** differences also exist.
- A combination of economic strength, healthcare and investment could influence the density of medical and nursing personnel across regions

Health expenditure and Medical doctors density, Coloured by Region



Health expenditure and Nursing and midwifery density, Coloured by Region





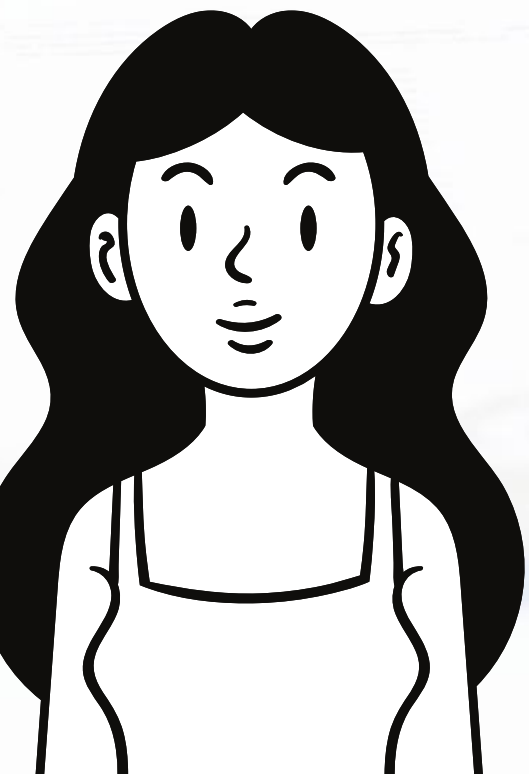
Other Impacting Variables



How does **tertiary education** differ throughout different **countries** and does it relate to the **density of medical staff**?

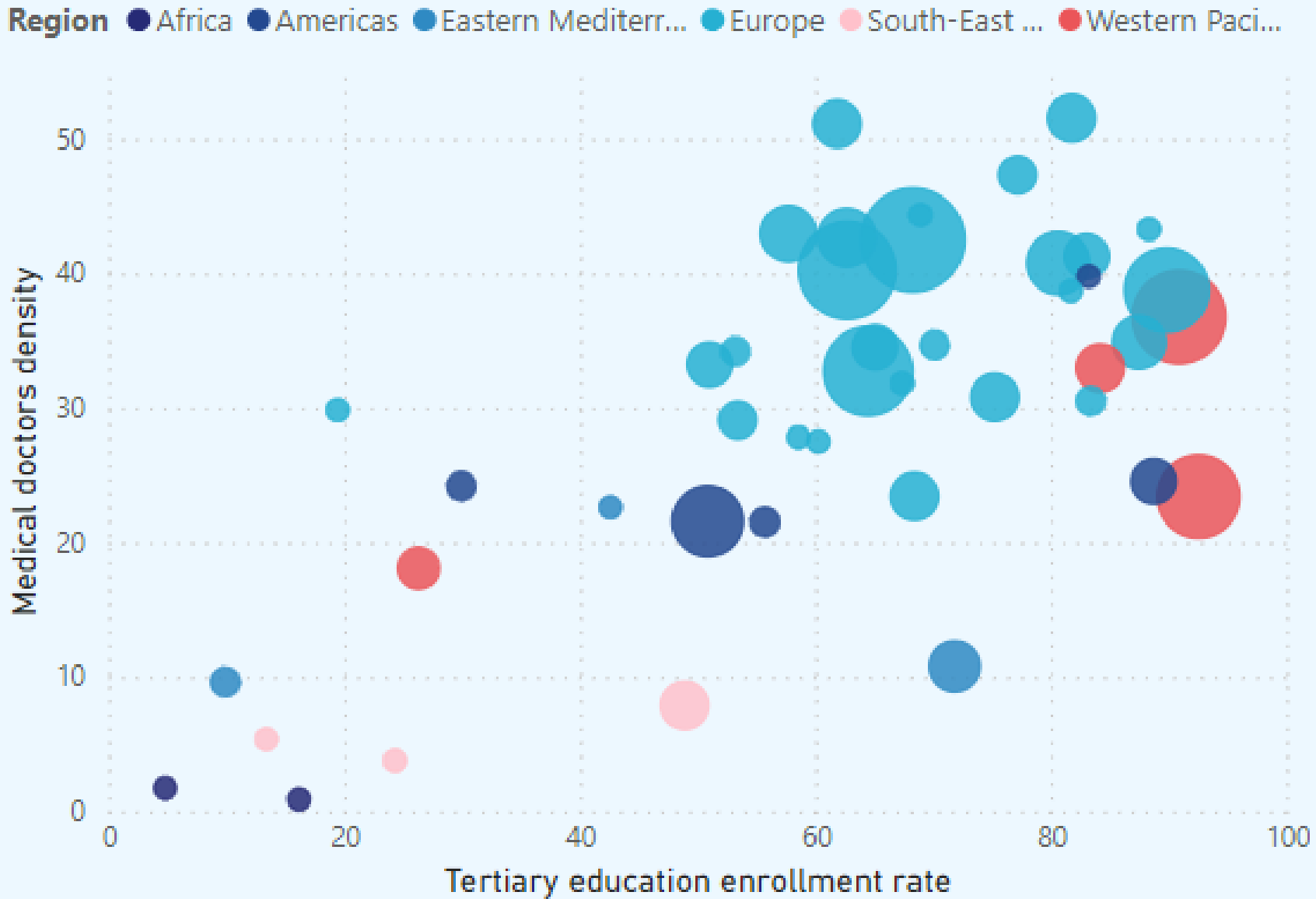


Methodology: Correlation analysis

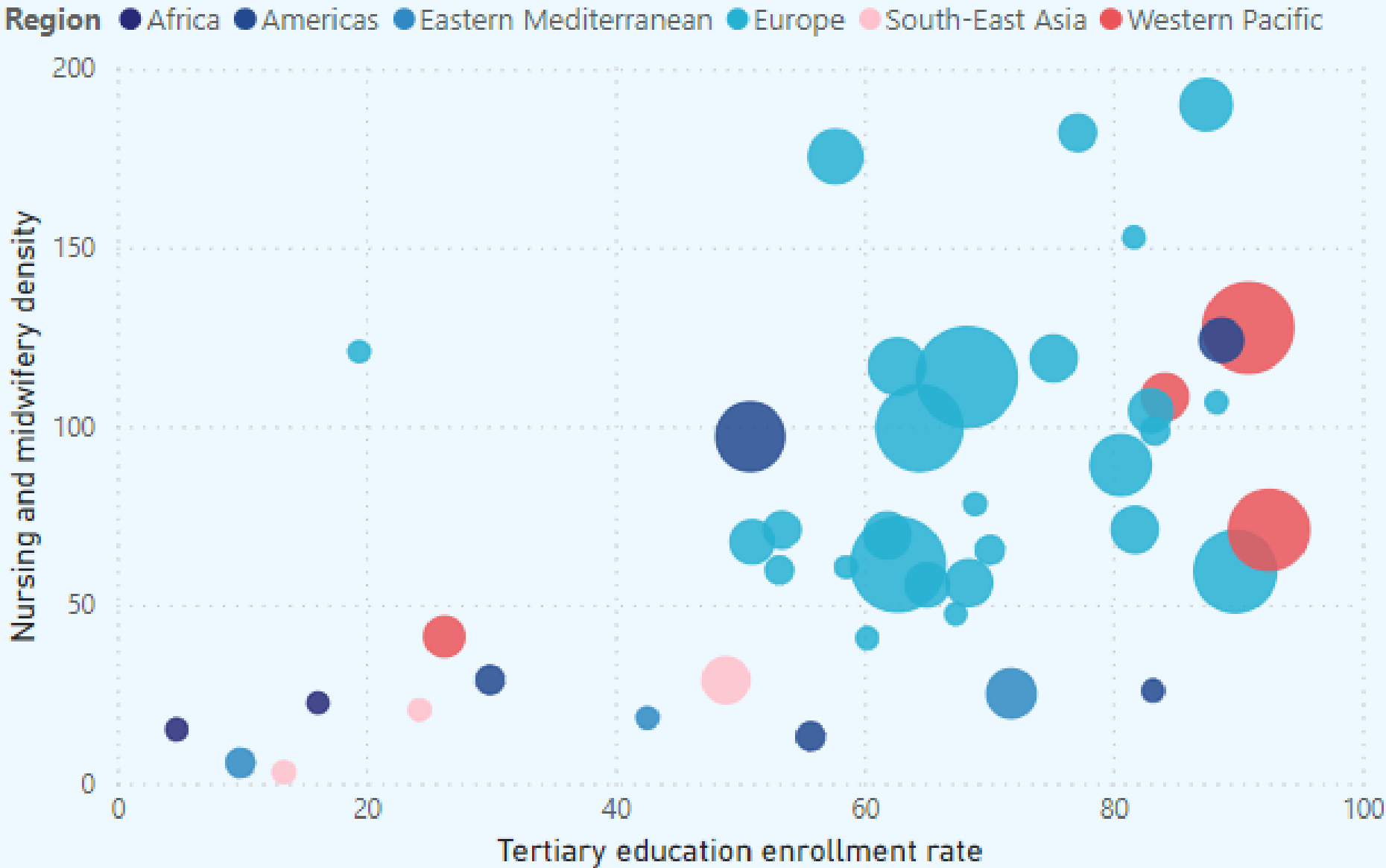


How does **tertiary education enrollment** differ throughout different **regions** and does it relate to the **density of medical staff**?

Tertiary Enrollment Rate compared to Medical Doctor Density, Sized by University Count per Country

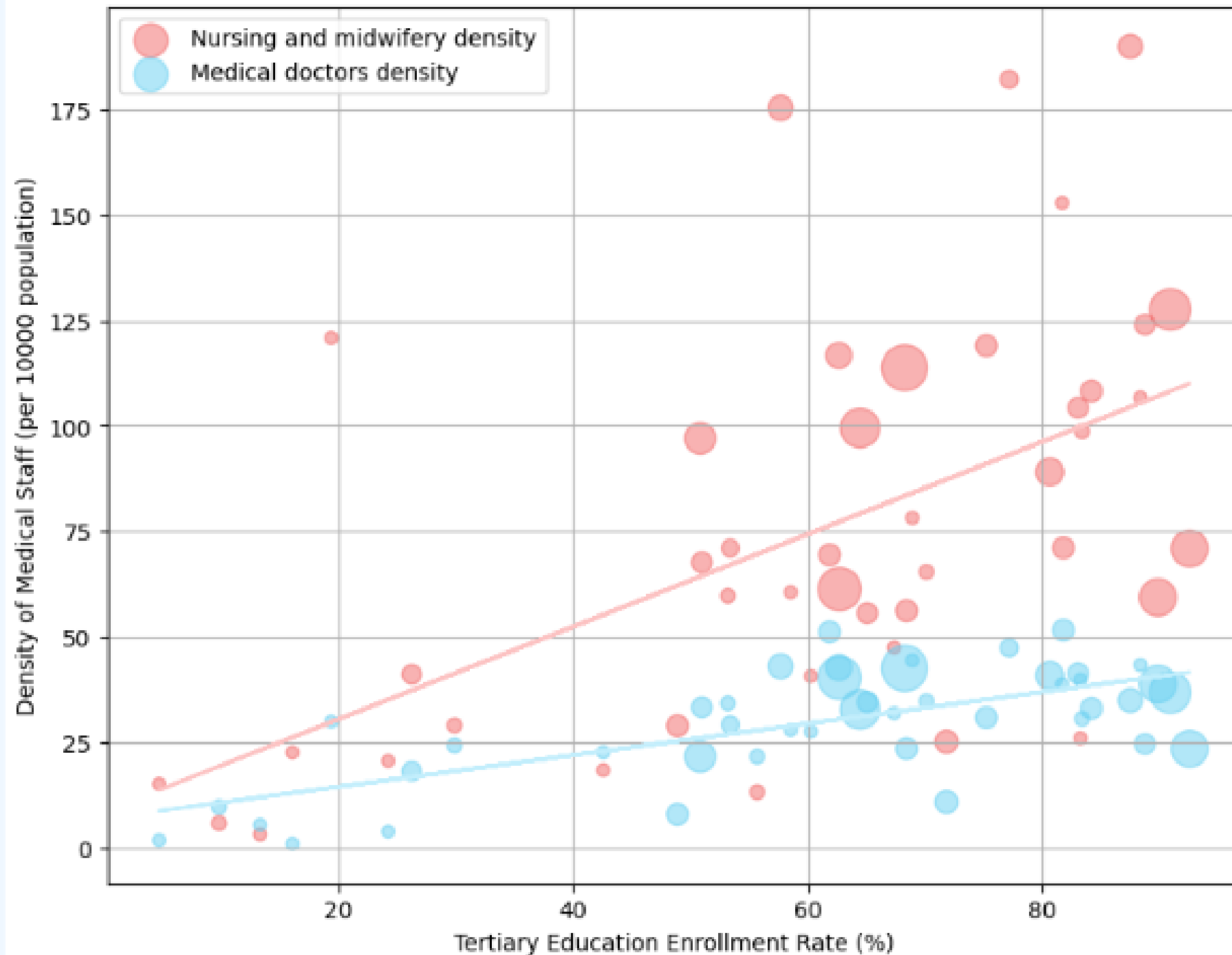


Tertiary Enrollment Rate compared to Nursing and midwifery density, Sized by University Count per Country

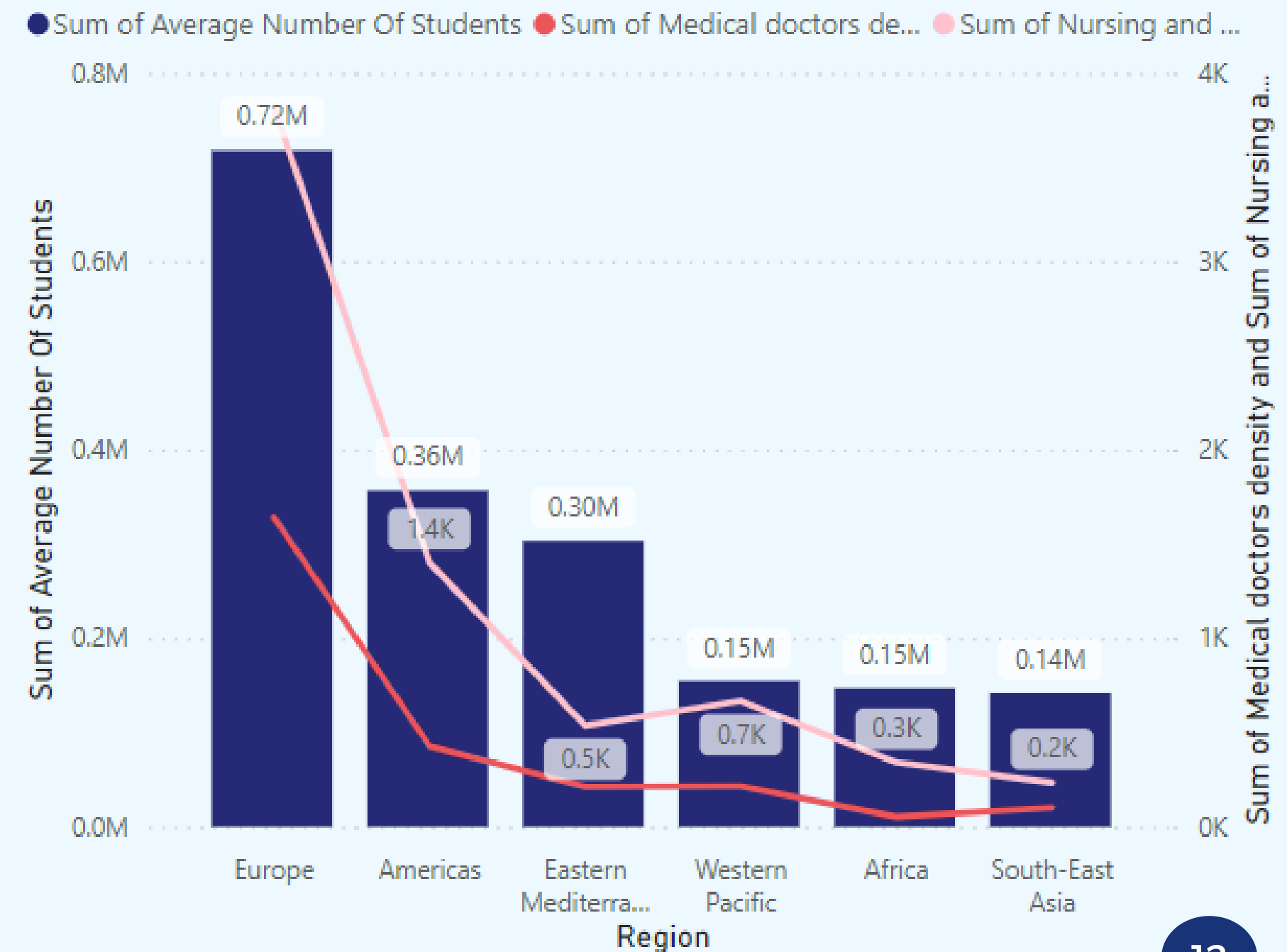


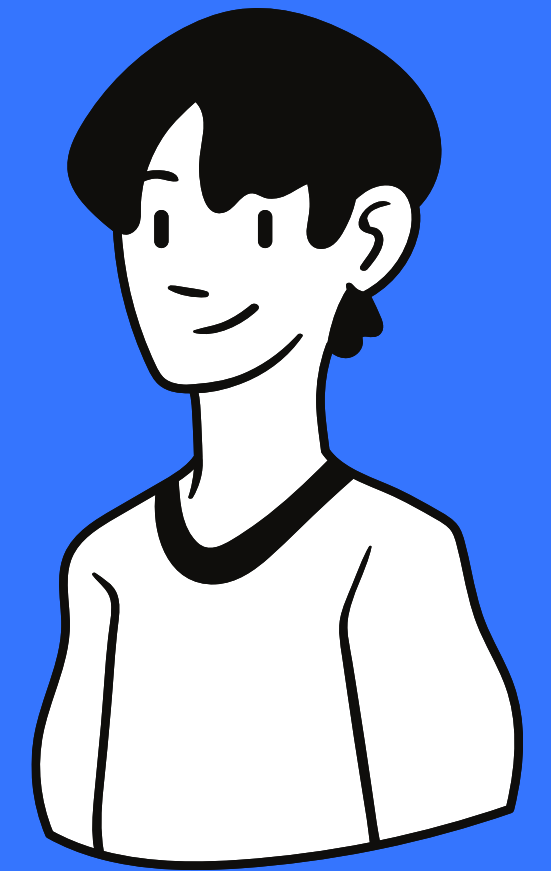
How does the **density of medical staff** differ in terms of **tertiary education**?

Correlation of Tertiary Education Enrollment with Medical Staff Density



Sum of Average Number Of Students, Sum of Medical doctors density and Sum of Nursing and midwifery density by Region





Address Limitation



- Our dataset only includes data for the year 2017, and it represents aggregated annual data without any time hierarchical structure.
- As a result, we are unable to perform analyses at smaller time intervals, such as by day or month.
- This limitation restricts the scope of our analysis and prevents us from continuous monitoring and tracking within the year.





Tracking Health and SDG Progress Through Key Indicators

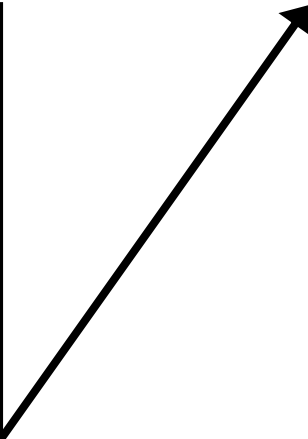


- ✓ Anomaly Detection and Swift Action
- ✓ Facilitating Timely Problem Analysis
- ✓ Continuous Progress Evaluation



Indicator	Number	Timestamp
Communicable disease deaths this year	11,727,35	2024-11-25 18:28:33
Seasonal flu deaths this year	461,365	2024-11-25 18:28:33
Public Education expenditure today	8,844,374,055	2024-11-25 18:28:33

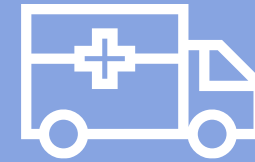
worldometer		Population	Coronavirus
WORLD POPULATION			
8,190,215,260	Current World Population	[+]	
119,628,129	Births this year	[+]	
280,405	Births today	[+]	
56,369,335	Deaths this year		
132,128	Deaths today		
63,258,794	Net population growth this year	[+]	
148,277	Net population growth today		
GOVERNMENT & ECONOMICS			
\$ 13,406,459,912	Public Healthcare expenditure today	[+]	
\$ 8,879,580,248	Public Education expenditure today	[+]	



Call to Action



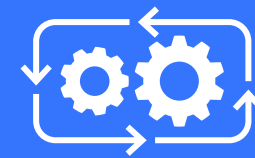
- Address regional disparities in healthcare staffing and access.
- Align health workforce planning with broader economic strategies to improve life expectancy and health outcomes.
- Strengthen tertiary education systems to train more healthcare professionals.
- Systematically tracking progress and addressing issues by escalating them to appropriate levels of authority when necessary.



**Promote
Equitable
Resource
Distribution**



**Invest in
Education**



**Integrate
Economic
Development and
Health Policies**



**Monitoring and
Escalation**





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+
**Thank
You.**

For Your Attention

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References

Data Sourcing:

- Country Statistics - UNData: <https://www.kaggle.com/datasets/sudalairajkumar/undata-country-profiles>
- Global Health Workforce statistics database: <https://www.who.int/data/gho/data/themes/topics/health-workforce>
- World University Rankings: <https://www.kaggle.com/datasets/mylesoneill/world-university-rankings>
- Worldometer - real time world statistics: https://www.worldometers.info/#google_vignette

Reference:

- Adebayo, T. S., Nwosu, L. C., Alhassan, G. N., Uzun, B., Özkan, O., & Awosusi, A. A. (2024). Effects of health expenditure, death rate, and infant mortality rate on life expectancy: A case study of the United States. *Energy & Environment*, 0958305X241281804
- World Health Organization. (2017). World health statistics 2017: Monitoring health for the SDGs, sustainable development goals. World Health Organization. <https://www.who.int/publications/i/item/9789241565486>

Appendix

Question 1-3: Full Data Frame

Region	Country	Nursing and midwifery density	Medical doctors density	Population density	Urban population	GDP per capita	Infant mortality rate	Health expenditure	Education government expenditure	Tertiary education enrollment rate	Unemployment (% of labour force)	Female life expectancy	Male life expectancy	Count Of Universities	Average Number Of Students
Eastern Mediterranean	Afghanistan	1.787	2.453	54.4	26.7	623.2	68.6	8.2	3.3	8.5	8.6	63.5	61		
Africa	Kenya	10.862		87.3	25.6	1376.7	39.4	5.7	5.3		10.8	67.8	63	1	72676
Africa	Mauritania	10.988	1.901	4.3	59.9	1235	68	3.8	2.9	5.35	12	64.1	61.2		
Americas	Canada	100.393		4	81.8	43205.6	4.7	10.4	5.3		7.1	83.8	79.7	25	28341
Europe	Czech Republic	101.132		137.5	73	17561.7	2.5	7.4	4.1	65.25	3.9	81.2	75.1	9	23694
Europe	Denmark	104.309	41.293	135.1	87.7	53149.3	3.5	10.8	8.6	83	6	82.2	78.1	6	22210
Europe	Belarus	106.701	43.303	46.7	76.7	5750.8	3.6	5.7	4.9	88.3	0.5	77.7	66.5	1	29303
Western Pacific	New Zealand	108.263	32.986	17.9	86.3	38294.3	4.4	11	6.4	84.15	5.5	83.1	79.5	7	17871
Americas	Dominican Republic	11.245													
Europe	Netherlands	111.251	35.875											13	17713
Europe	Germany	113.768	42.505	235.6	75.3	41686.2	3.4	11.3	5	68.25	4.2	82.9	77.9	39	28327
Europe	Sweden	116.712	42.69	24.2	85.8	50687.5	2.4	11.9	7.7	62.65	7.3	83.7	80	11	16667
Europe	Belgium	119.024	30.804	377.5	97.9	40277.8	3.5	10.6	6.6	75.2	8.3	83	78	7	23819
Western Pacific	Lao People's Democratic Republic	12.028	4.413	29.7	38.6	1850.2	47.3	1.9	3.3	16.9	1.5	66.8	63.9		
Africa	Cabo Verde	12.177		135.6	65.5	3080.2	22.5	4.8	5	21.65	10.5	74	70.1		
Americas	Guatemala	12.269	2.857	157.8	51.6	3903.5	26.9	6.2	3	21.85	2.4	75.6	69.2		

Appendix

Question 4: Full Data Frame

	Number	Indicator	Timestamp
0	11,727,355	Communicable disease deaths this year	25/11/2024 18:28
1	461,365	Seasonal flu deaths this year	25/11/2024 18:28
2	6,866,600	Deaths of children under 5 this year	25/11/2024 18:28
3	40,690,464	Abortions this year	25/11/2024 18:28
4	279,223	Deaths of mothers during birth this year	25/11/2024 18:28
5	45,965,354	HIV/AIDS infected people	25/11/2024 18:28
6	1,518,629	Deaths caused by HIV/AIDS this year	25/11/2024 18:28
7	7,419,348	Deaths caused by cancer this year	25/11/2024 18:28
8	356,237	Deaths caused by malaria this year	25/11/2024 18:28
9	11,674,961,357	Cigarettes smoked today	25/11/2024 18:28
10	4,516,001	Deaths caused by smoking this year	25/11/2024 18:28
11	2,259,425	Deaths caused by alcohol this year	25/11/2024 18:28
12	968,732	Suicides this year	25/11/2024 18:28
13	361,394,039,806	Money spent on illegal drugs this year	25/11/2024 18:28
14	1,219,463	Road traffic accident fatalities this year	25/11/2024 18:28
15	13,353,294,279	Public Healthcare expenditure today	25/11/2024 18:28
16	8,844,374,055	Public Education expenditure today	25/11/2024 18:28